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PUBLICATIONS

H-index : 4 (according to Google Scholar)

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Following the Hardy-Littlewood rule, the list of authors in most of my publications is sorted alphabetically.

Conference Proceedings

- Acharya, Anasuya, Carmit Hazay, and Rahul Satish (2026). "The Landscape of Reusable Garbling". In: vol. 2026, p. 492. URL: <https://eprint.iacr.org/2026/492>.
- Acharya, Anasuya, Karen Azari, Mirza Ahad Baig, Dennis Hofheinz, and Chethan Kamath (2025). "Securely Instantiating 'Half Gates' Garbling in the Standard Model". In: PKC 2025, May 12-15, 2025. Røros, Norway: Springer, pp. 37-75.
- Acharya, Anasuya, Karen Azari, and Chethan Kamath (2025). "On the Adaptive Security of Free-XOR-Based Garbling Schemes in the Plain Model". In: EUROCRYPT 2025, May 4-8, 2025. Madrid, Spain: Springer, pp. 214-244.
- Acharya, Anasuya, Carmit Hazay, Vladimir Kolesnikov, and Manoj Prabhakaran (2025). "Towards Building Efficient SCALES Protocols". In: ASIACRYPT 2025, December 8-12, 2025. Melbourne, Australia: Springer.
- Acharya, Anasuya, Carmit Hazay, and Muthuramakrishnan Venkatasubramanian (2025). "On Achieving 'Best-in-the-Multiverse' MPC". In: TCC 2025, December 1-5, 2025. Aarhus, Denmark: Springer.
- Acharya, Anasuya, Carmit Hazay, Vladimir Kolesnikov, and Manoj Prabhakaran (2024). "Malicious Security for SCALES - Outsourced Computation with Ephemeral Servers". In: CRYPTO 2024, August 18-22, 2024. Santa Barbara, USA: Springer, pp. 3-38.
- Acharya, Anasuya, Tomer Ashur, Efrat Cohen, Carmit Hazay, and Avishay Yanai (2023). "A New Approach to Garbled Circuits". In: ACNS 2023, June 19-22. Kyoto, Japan: Springer, pp. 611-641.
- Acharya, Anasuya, Carmit Hazay, Oxana Poburinnaya, and Muthuramakrishnan Venkatasubramanian (2023). "Best of Both Worlds - Revisiting the Spymasters Double Agent Problem". In: CRYPTO 2023, August 20-24, 2023. Santa Barbara, USA: Springer, pp. 328-359.
- Acharya, Anasuya, Carmit Hazay, Vladimir Kolesnikov, and Manoj Prabhakaran (2022). "SCALES - MPC with Small Clients and Larger Ephemeral Servers". In: TCC 2022, November 7-10, 2022. Chicago, USA: Springer, pp. 502-531.
- Prabhakaran, Manoj M., Anasuya Acharya, and Akash Trehan (2019). "An Introduction to the CellTree Paradigm". In: ICIS 2019, Volume 11952 LNCS, 16-20 December 2019. Hyderabad, India: Springer.
- Sanyashi, Tikaram, Anasuya Acharya, and Bernard Menezes (2019). "Plaintext Recovery Attacks and their Mitigation in an Application Specific SHE Scheme". In: PDCAT 2019, 5-7 December 2019. Gold Coast, Australia: IEEE Xplore.
- Patil, Vishwas, Anasuya Acharya, and R. K. Shyamasundar (2018). "Landcoin: A Land Management System using Litecoin Blockchain Protocol". In: SDLT3 2018, 12 November 2018. Gold Coast, Australia.

EDUCATION

Ph.D. in Computer Engineering

Bar-Ilan University

July 2025

Ramat Gan, Israel

CGPA: 86.94 / 100

M.Tech. in Computer Science and Engineering (CSE)

IIT Bombay

July 2020

Mumbai, India

CGPA: 9.54 / 10

B.Tech. in CSE

NIT Trichy

May 2017

Tiruchirappalli, India

CGPA: 8.06 / 10

REFEREES

Prof. Claudio Orlandi

Aarhus University

Aarhus, Denmark

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Postdoc Host

Prof. Carmit Hazay

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Ramat Gan, Israel

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PhD Thesis Supervisor

Prof. Manoj M. Prabhakaran

IIT Bombay

Mumbai, India

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www.cse.iitb.ac.in/~mp/

Masters Thesis Supervisor

SERVICE

Workshop Talks

- *Maliciously Secure SCALES Protocols* (2024). Cryptography in the Blockchain Era (CIBE) 2024, 23-27 June 2024, Bertinoro, Italy.
- *Maliciously Secure SCALES Protocols* (2024). Theory and Practice of Multi-Party Computation Workshop (TPMPC) 2024, 3-6 June 2024, TU Darmstadt, Germany.
- *SCALES: MPC with Small Clients and Larger Ephemeral Servers* (2023). ACE Conference (Algorand Centres of Excellence) 2023, 10-12 January 2023, Hotel Arts, Barcelona.
- *SCALES: MPC with Small Clients and Larger Ephemeral Servers* (2022). Theory and Practice of Multi-Party Computation Workshop (TPMPC) 2022, 7-10 June 2022, Aarhus University, Denmark.
- *CellTree: A new Paradigm for Distributed Data Repositories* (2019). Talk, 28 June 2019, IIT Bombay, Mumbai, India.

Organizing

- 39th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (2019). 11-13 December 2019, IIT Bombay, Mumbai, India.
- ACM India Grad Cohort 2018 (2018). 6-7 July 2018, IIT Bombay, Mumbai, India.

EXPERIENCE

Postdoctoral Researcher

Aarhus University

March 2025 – July 2026 Aarhus, Denmark

Postdoctoral Researcher in secure multiparty computation (MPC) hosted by Prof. Claudio Orlandi in the Cryptography and Cybersecurity Section, Aarhus University.

Research Assistant

Information Security Research and Development Center (ISRDC)

July 2017 – July 2020 Mumbai, India

A three year scholarship student and Research Assistant studying distributed ledger technologies (DLT) in ISRDC, IITB.

Research Intern

Indian Statistical Institute

December 2015 – January 2016 Kolkata, India

Research intern in cryptography under Prof. Shubhamoy Maitra, studying and attempting cryptanalysis of the RC4 stream cipher.

REVIEWING

PC Member

TCC '26 ASIACRYPT '26

Sub-reviewing

CRYPTO '22, '24, '25, '26 ITC '25
EUROCRYPT '22, '24, '25 TCC '22, '24
ASIACRYPT '25 PODC '26
PKC '26

ACHIEVEMENTS

GATE (Graduate Aptitude Test for Engineering)
Score of 791 and AIR (All India Rank) 231 in the GATE Computer Science and Information Technology paper in February 2017

AP (Advanced Placement Exam)
Scored 3/5 in Physics C Mechanics, Computer Science A, Calculus AB in May 2012

AAT (National Award of Academic Aptitude and Achievement Test)
Scored 99 percentile in quantitative aptitude in May 2007

SOFTWARE SKILL SET

LaTeX GIT Linux(Ubuntu)
Java C C++ SEBOL JavaScript
Python PHP MATLAB

LANGUAGES

English
Hindi
Bengali
Japanese
Mandarin
Bahasa Indonesia
Danish

EXTRA-CURRICULAR

JLPT N2
HSK 3
TEFL
BMC @ HMI

HOBBIES

Trekking Learning Languages
Writing Crash Coursing Visual Arts